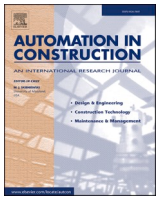
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A digital monitoring, delay detection and visualisation framework for construction projects: RealCONs

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ABSTRACT

Accurate and resilient monitoring of construction projects remains challenging due to fragmented reporting, data uncertainty and delayed system integration. This paper evaluates RealCONs, a QR-enabled real-time monitoring framework that integrates BIM, mobile scanning, cloud-based SQL storage, and Power BI analytics to support live project control. A 90-day comparative case analysis of two concurrent Electrical and Instrumentation projects benchmarked RealCONs against a conventional tracking system. Performance was assessed using Earned Value and Earned Schedule metrics, supported by Chi-square and two-proportion tests, confidence intervals, normality testing, regression forecasting, and non-parametric Wilcoxon and Mann–Whitney analyses. Data continuity strongly favoured RealCONs, with five missing earned-value days compared with 35 in the comparator project ($\chi^2 = 28.93$, $p < .001$). Across 51 paired days, RealCONs achieved superior CPI (1.02 vs 0.90) and SPI (1.01 vs 0.89). During a delay event (Days 33–37), RealCONs maintained measurable progress and statistically significant SPI predictability, while the comparator recorded zero earned value. Overall, RealCONs enabled earlier delay detection, improved forecast reliability and scalable, real-time decision support aligned with Industry 4.0 objectives.

1. Introduction

Delays in large-scale construction projects remain a persistent challenge, particularly within Electrical and Instrumentation (E&I) services such as cable tray installation, cabling, and testing and commissioning [1–5]. These activities are highly vulnerable to trade interdependencies, sequencing constraints, and limited work fronts, with disruptions often cascading across the programme. As E&I tasks are typically situated on the critical path, their timely completion is essential to ensure power availability for commissioning and handover [6]. Consequently, rigorous monitoring and early issue identification are crucial, especially given the limited opportunities for trade overlap [7]. In practice, however, E&I activities are prone to further delays due to incomplete prerequisite works, site congestion, and sequencing misalignments [8–11]. Traditional progress reporting methods, which are primarily reliant on manual inputs and periodic updates, are time-consuming and prone to errors, thereby reducing opportunities for timely corrective action [11–13]. To improve visibility, automated tracking solutions such as laser scanning, RFID, photogrammetry, ultra-wideband (UWB), and wireless sensor networks (WSNs) have been explored [14–20]. While these systems can provide accurate and frequent progress updates [21–23], their high costs, training requirements, and operational complexity limit scalability, particularly in the dynamic, delay-sensitive context of E&I works [24–26].

Mainstream project management platforms such as Primavera P6 and MS Project remain central to baseline development, look-ahead planning, and critical path tracking [27–29]. Yet, they are heavily dependent on retrospective manual updates. BIM-enabled 4D scheduling has improved the visualisation of time–space conflicts and sequencing constraints [9,30,31], but its integration with real-time field data remains underdeveloped [32–39]. Similarly, Earned Value and Earned Schedule Management provide structured metrics for performance assessment [40–44], yet delayed or inaccurate data inputs undermine their usefulness. As digital workflows gain momentum across the industry, there is a clear need to connect field-level operations with planning platforms through integrated, real-time monitoring, particularly for E&I activities where delays are both frequent and consequential. This paper addresses this gap by proposing the RealCONs framework, which integrates QR code-based mobile scanning, a cloud-based SQL database, and dynamic analytics via Power BI. RealCONs

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creates a closed feedback loop between site-level data, BIM models, schedule baselines, and cost structures, enabling real-time earned value tracking and proactive delay management. The framework advances current practice in three key respects:

1) It integrates existing digital tools into a process-based workflow, reducing the need for additional investment in new systems.
2) It employs BIM-enabled QR code localisation, ensuring precise mapping of on-site progress to planned tasks.
3) It delivers live delay detection and impact forecasting, empowering project managers to act proactively rather than retrospectively.

However, the research is guided by five objectives:

1) Establish BIM-linked QR code localisation to connect construction tasks to their digital representations.
2) Develop an end-user mobile interface for immediate, on-site data capture and entry.
3) Integrate the mobile application with a cloud-based SQL database for centralised data storage and retrieval.
4) Design a data model for Power BI analytics to evaluate performance using real-time EVM metrics.
5) Validate the framework on real-world E&I projects to assess applicability and impact.

The RealCONs system combines widely available technologies: BIM (Revit 2023) for digital representation, SQL Server (version 16, 2022) for structured data storage, smartphone-enabled QR scanning (Android OS 12+ and iOS 15+) for live field input, and Power BI (version 2.145) for interactive dashboards. Together, these components support the early detection of delays, evaluation of their impact on cost and time, and proactive project management. Its low-cost, scalable architecture is tailored to the dynamic nature of E&I activities, offering improved coordination and disruption management. The framework was developed following a literature-based gap analysis, built within Power BI, and validated on multiple large-scale E&I projects. The findings confirm its feasibility and demonstrate its potential to support proactive, data-driven decision-making.

2. Research background

This section outlines the technological and methodological foundations of the proposed framework. It reviews the challenges in Electrical and Instrumentation (E&I) work, recent developments in real-time tracking, the use of smartphones for on-site data capture, and the potential of visual analytics tools, such as Power BI, to support timely project decision-making.

2.1. Smartphone-based QR tracking for on-site progress monitoring

Recent developments in Industry 4.0 and mobile computing highlight smartphones as practical tools for construction information management [45,46]. Mobile applications are now widely used for document control, progress updates, and project tracking [47–53]. Leveraging built-in sensors and connectivity (e.g., Bluetooth, 3G/4G), smartphones can capture and transmit data across large project areas [54–57]. When combined with QR codes, smartphones offer a cost-effective and accessible means of real-time monitoring. QR codes placed at installation points allow site crews to record completion data, which is synchronised to cloud databases for centralised analysis [56,58,59]. This approach reduces documentation errors, improves coordination, and enhances efficiency. While several studies have highlighted the potential of QR-based tracking, the systematic use of this technology to support structured E&I data capture and integration into live analytical dashboards remains underdeveloped [7,26]. This research builds upon these foundations to establish a more comprehensive, data-driven solution.

2.2. Visual analytics and performance monitoring using power BI

Descriptive and graphical analytics remain the most widely used approaches to monitor construction progress and performance [60,61]. These methods provide retrospective insights into “what happened” by visualising historical and real-time data trends. When paired with business intelligence tools such as Power BI, these insights can be presented in interactive dashboards, enabling the rapid identification of deviations, supporting informed decision-making, and enhancing communication among stakeholders [62–67]. Power BI is particularly effective for visualising key performance metrics, such as the SPI and CPI, due to its seamless integration with the Microsoft ecosystem and its support for large-scale, cloud-based collaboration [68–71]. Its intuitive interface and analytical features (e.g. DAX, Power Query) make it accessible to both technical and non-technical users, including site managers and engineers [68,69,72–74]. Compared with other BI tools, Power BI is cost-effective, scalable, and suitable for real-time construction monitoring [75].

Despite widespread use of these tools individually, the literature highlights a lack of integrated, field-deployable frameworks that combine real-time data capture with performance analytics—especially within E&I scopes. Previous studies have not adequately addressed the modelling of data flow between project entities (e.g. contractors, engineers, consultants), nor the relationships between automated tracking systems and early-stage delay notification mechanisms [76–78]. This paper addresses this gap by proposing a comprehensive framework that unifies QR-based mobile tracking, cloud-based data integration, and Power BI-driven analytics for improved delay detection and proactive project control.

2.3. Comparative contribution of realcons

To address these gaps, this paper introduces the RealCONs framework, which unifies QR-based smartphone tracking, SQL-based data integration, and Power BI analytics. The framework offers three key contributions: (1) **process-based integration** of widely available digital tools into a coherent workflow, reducing the need for additional investment; (2) **BIM-enabled QR localisation**, improving precision in connecting field data to planned tasks; and (3) **real-time earned value forecasting**, enabling proactive delay detection and early intervention. The unique contribution of RealCONs is highlighted in Table 1, which compares its features with those of leading approaches identified in the literature.

3. Research methodology

The research method is based on previous studies, described in detail in Section 2, on developing and testing digital technologies in project management, particularly in project tracking and visualisation. These studies have developed their own processes and prototypes for data collection, 3D presentation, and project timeline monitoring. However, they did not specifically focus on early delay warnings or automated analysis of their impacts on project performance. The method involves developing and testing a digital framework designed to instruct an integrated process for real-time data collection from construction sites and visualising project status. This includes performing specific tasks such as identifying the location of QR codes on site using BIM, integrating smartphones for QR code scanning, and connecting a cloud-based SQL database with Power BI to provide early notifications and analyse their impact using earned value (EV) indexes. Following its development, the testing involved case studies (e.g. Electrical and Instrumentation works), where real-time installation data and BIM component location codes were centralised to create a data model for presenting analytical reports. The experiment aimed to enable project managers and key decision-

Table 1
Comparative overview of previous studies and RealCONs framework.

Feature / capability	Previous studies [8,26,27,79,80]	Realcons framework (proposed)
Integration of digital tools	TYPICALLY FOCUS ON SINGLE TOOLS (E.G., BIM FOR VISUALISATION, RFID FOR TRACKING).	PROCESS-BASED INTEGRATION OF BIM, QR, PLANNING_DB, SQL, AND POWER BI INTO A UNIFIED WORKFLOW.
Real-time earned value management	LIMITED OR ABSENT; USUALLY BASED ON PERIODIC REPORTS.	LIVE DASHBOARDS WITH EVM METRICS (SPI, CPI) SUPPORTING EARLY DETECTION OF DISCREPANCIES.
BIM–QR code linkage	RARE; QR USED MAINLY FOR MATERIAL TRACKING.	NOVEL USE OF BIM FOR QR PLACEMENT TO DIRECTLY LINK SITE DATA WITH PLANNED ACTIVITIES.
Data model architecture	FRAGMENTED OR TOOL-SPECIFIC, OFTEN LACKING INTEROPERABILITY.	CENTRALISED SQL-BASED PROJECT_DB INTEGRATING DESIGN, SCHEDULE, SITE, AND ANALYTICS DATA.
Early warning and forecasting	RETROSPECTIVE AND REACTIVE IN MOST CASES.	PROACTIVE FORECASTING OF TIME AND COST IMPACTS FROM EARLY DISCREPANCIES.
Validation in E&I projects	RARELY TESTED IN COMPLEX, FAST-PACED DOMAINS.	APPLIED TO REAL-WORLD E&I PROJECTS, DEMONSTRATING ROBUSTNESS AND SCALABILITY.

makers to identify issues (e.g., delays) and make timely cost and schedule forecasts. For this purpose, the research methodology framework, illustrated in Fig. 1, comprises two main stages: the design and coding processes of the proposed framework. These are further divided into five steps: (3.1) framework design, outlining its structure; (3.2) QR code design; (3.3) smartphone application development and its integration with the database; (3.4) Power BI design; and (3.5) a case study for testing. Additionally, a statistical validation step (4) is included to assess the proposed framework using case studies.

3.1. Proposed framework design

3.1.1. RealCONs overview

To clarify the structure and interaction between components of the RealCONs framework, it is helpful to outline the proposed process on major construction sites, following the sequence of 1) real-time data collection, 2) data integration, and 3) data analysis and visualisation, as illustrated in Fig. 2.

The RealCONs framework was developed to enable real-time acquisition, integration, and analysis of construction site data. As shown in Fig. 3, its architecture is composed of three core and two supporting components, designed according to an object-oriented structure. Table 2 presents the core and supporting components.

The actionable implementation follows a staged integration pathway that begins with establishing QR-enabled data capture at the Project Entry Gate, continues with linking site inputs to structured scheduling data in Planning_DB (Primavera P6) and schedule of quantity (SoQ) data in BIM (Revit), and proceeds through the validation and merging of records in Project_DB (SQL Server) before transforming and presenting analytical outputs in Analytical_DB (Power BI). For projects equipped with BIM and P6 structures, the configuration effort remains modest, involving the setup of QR-generation scripts, API endpoints, authentication services, and ETL (Extract, Transform, Load) routines, all of which can be achieved through standard web technologies. Procurement and policy mechanisms such as mandated digital site records, API-ready contractor datasets, and contractual data-quality requirements further support consistent adoption across the supply chain.

Within this architecture, two functional domains operate together. The Project Entry Gate governs the generation and issuance of QR codes from predefined elements in the Project Database, informed by BIM components and P6 logic. At the same time, the Project Management domain captures real-time site progress through QR scans, transmits the data to the Project Database for consolidation, and channels it onward to the Analytical Database for visualisation of progress status, performance trends, and earned value indicators. Data integrity and system security are maintained through multi-layered controls, including credential-based authentication, encrypted transmission, IP-based trust verification, and role-based access. Users scanning a QR code are directed to a secure login portal, and all device-to-database communication is handled through standard web protocols (HTML forms, PHP/APIs, MySQL/SQL), supported by a cross-platform interface to ensure operational usability. Reproducibility and robustness are enhanced through explicit error handling and offline capabilities. Each scan undergoes local validation before a payload is constructed; if connectivity is unavailable, records are queued offline and synchronised automatically once the network is restored. Server-side checks enforce schema compliance, duplication control, and business-rule adherence, returning standard status codes. All site updates are logged in SQL Server and integrated with BIM-derived quantities and the relevant Primavera P6 activities, WBS (Work Breakdown Structure) and CBS (Cost Breakdown Structure) codes. Analytical_DB then refreshes earned value and earned schedule metrics (CPI, SPI, SVc, SPIc) and issues warnings where baseline data are incomplete, enhancing transparency and supporting proactive control. Table 3 describes the lifecycle mapping approach.

However, Fig. 4 illustrates this closed-loop workflow, from QR scan and validation through secure transmission, conflict handling, offline recovery, and downstream analytics. This demonstrates how design data

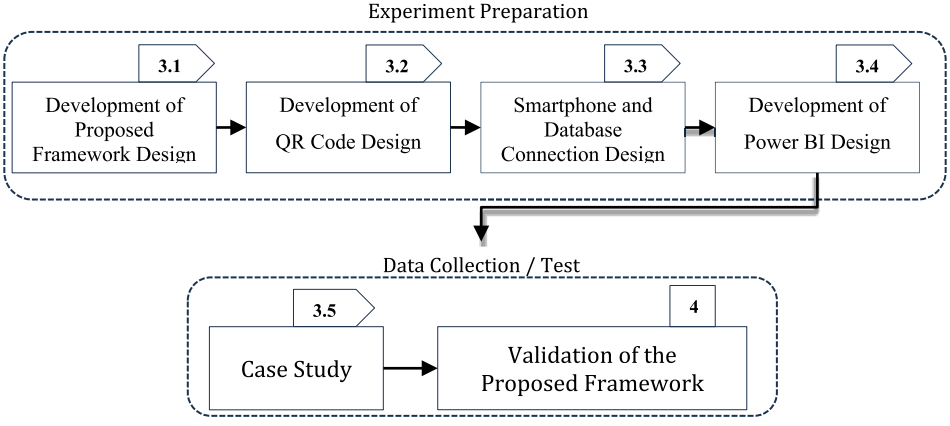


Fig. 1. Research method framework.

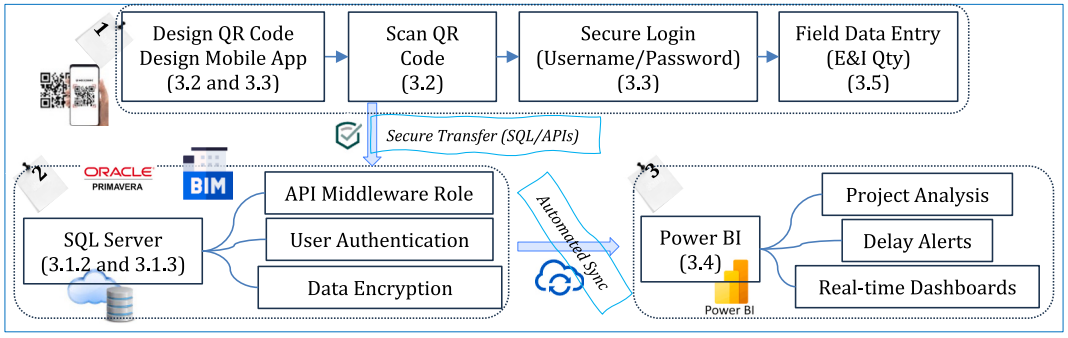


Fig. 2. Research conceptual model.

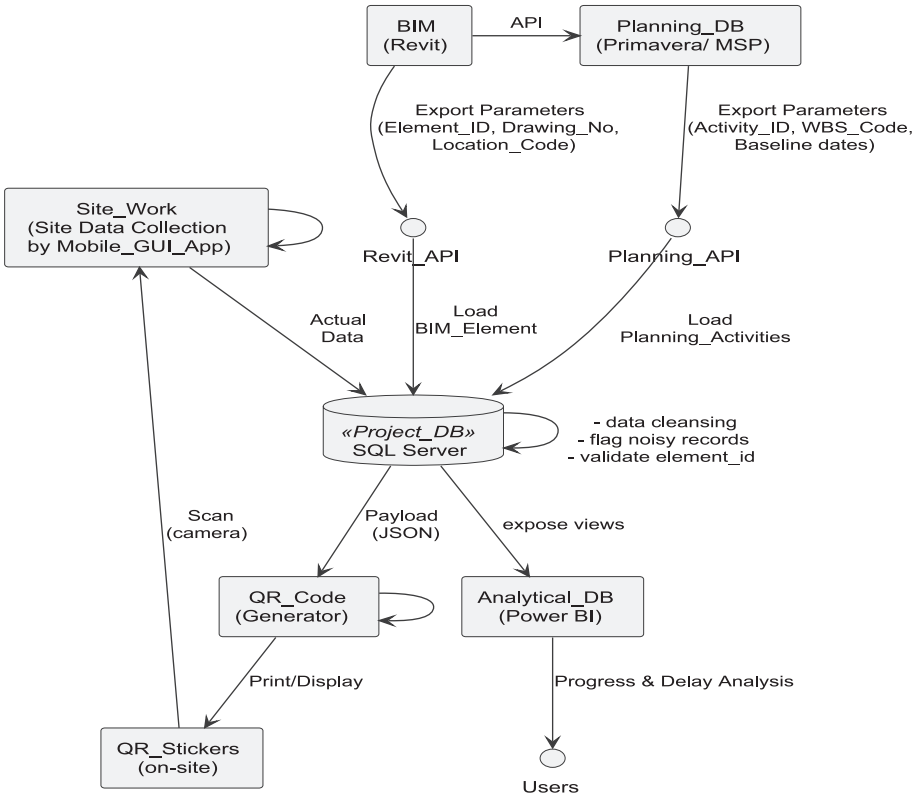


Fig. 3. RealCONs framework and its classes.

Table 2
RealCONs framework: Core and supporting components.

Type	Components	Description
Core	Site_Work	The front-end module is responsible for on-site data capture via QR code scanning, recording user credentials, installation quantities (e.g., cables, trays), building location codes, and issue-related comments.
	Project_DB	A centralised SQL Server database that functions as the backbone of the system, storing all project-related records
	Analytical_DB	A Power BI–linked platform facilitating project analytics, including earned value (EV) metrics
	Planning_DB	A repository for planned schedules (WBS) and cost breakdowns (CBS), derived from Primavera P6
Supporting	BIM	A 3D environment used to quantify associated installation materials and identify the spatial location of QR codes

defines QR elements, planning data aligns WBS and durations, site data populates progress records, and analytics convert them into actionable insights. Collectively, this integrated approach enhances accuracy, reduces manual effort, and facilitates early detection of delays. All scripts, configurations, Power Query transformations, and DAX measures are included in Appendix 1.

3.1.2. *Database schema and relationships*

To enable integration between BIM data, schedule data, and field-scanned progress records, a relational schema was designed in SQL Server (Project_DB). This revised schema (presented in Fig. 3, adapted from the workflow diagram) consolidates three key streams of information:

- BIM data from Revit (element IDs, geometry references, and quantities),
- Schedule data from Primavera/MSP (WBS codes, activity IDs, planned dates, and resources),

Table 3
Lifecycle mapping.

Stage	Inputs	Framework Activity	Outputs
1. Establishment	BIM model (Revit): quantities (SoQ)	Define elements; generate IDs for QR codes	Structured element list in Project_DB
2. Planning (P6)	WBS, Durations, CBS codes	Link BIM elements to schedule; align SoQ with activities	Integrated BIM–Primavera P6 dataset for QR generation
3. Entry Gate	Project_DB records; QR scripts	Generate and issue QR codes; set capture rules	QR-enabled workflow with access controls
4. Site Capture	Mobile scans; user credentials	Validate locally; submit or queue offline	Clean progress records sent to Project_DB
5. Server Processing	SQL Server records; validation rules	Merge, de-duplicate, and enforce business rules	Authoritative progress dataset mapped to BIM and P6
6. Analytics	Consolidated records	ETL to Power BI; refresh EV and ES (CPI, SPI, SVc, SPIc)	Dashboards and alerts on delays or missing baselines
7. Management Action	Power BI insights	Assess deviations; adjust plans	Updated forecasts and feedback into the workflow

- Real-time site data captured via mobile QR scanning (progress quantities, timestamps, and resource hours).
- Data integrity is ensured through primary and foreign key relationships. The tables are linked as follows (Table 4):

As seen in Fig. 5, these one-to-many relationships allow a single BIM element to be associated with multiple scheduled tasks and progress entries. The schema supports real-time queries by analytical tools such as Power BI.

3.1.3. BIM data extraction and integration workflow

The RealCONs framework requires structured extraction of BIM data to ensure compatibility with both Planning_DB and site-collected progress data. Revit models were used as the primary BIM, and data were exported via API scripts into the Project_DB.

- **Data Extracted from BIM:**
  - **Element IDs** (unique identifiers for each physical component).
  - **Spatial coordinates** (positions of components within the model).
  - **Quantities** (e.g., length of cable tray and cables in metres, volume of fittings).
  - **System classification codes** (e.g., electrical, mechanical, architectural).
- **Integration with Project Data:**
  - Each BIM Element ID is mapped to a WBS activity via a lookup table.
  - A corresponding QR code is generated that encodes the element ID, drawing reference, and WBS code.
  - On-site, the QR sticker is affixed to the physical element, enabling direct progress capture through mobile scanning.
- **Linking to Schedule Activities:**
  - WBS codes from the Planning_DB are cross-referenced with BIM element IDs.
  - This ensures that scanned progress updates (e.g., 30 m of tray installed) are automatically matched to the correct scheduled task in Project_DB.
  - The linkage is many-to-one: multiple BIM elements (e.g., tray segments) can feed into a single WBS activity (e.g., “Install cable tray in Zone A").
- **Data Flow:**
  - **Extraction**: The Revit API exports element IDs, quantities, and attributes.
  - **Transformation**: Data stored in Project_DB with WBS lookups.
  - **Integration**: QR codes generated and attached to elements on-site.
  - **Execution**: Field scans update progress in real time.

This workflow ensures seamless traceability, which is central to RealCONs real-time monitoring capability.

3.2. QR code design

From Figs. 3 and 5, the QR code generation sequence: BIM model data flows into a centralised SQL Server acting as the primary integration hub. Automated APIs extract element-specific attributes, including spatial locations, installation quantities, building codes, and drawing numbers, directly from the 3D model, ensuring digital-physical consistency. These datasets undergo programmatic encoding via Python scripts to embed critical metadata (BIM element identifier, building code, drawing number, location tag) within QR codes, which are subsequently printed and affixed to corresponding on-site locations (Tables 5 and 6). This process bridges the gap between digital design and physical implementation, enabling precise spatial tracking and efficient field data capture.

Figs. 6–8 demonstrate the project's 3D model and construction site real perspectives, drawn from a case study that validates the framework. However, for confidentiality compliance with case study agreements, actual QR codes captured during field implementation have been systematically obscured in Figs. 7 and 8. Fig. 6 presents an annotated BIM snippet illustrating unique tag numbers. Notably, QR code symbols, which are absent in the original 3D model, have been superimposed onto this visualisation. These symbols serve as digital references for the strategic placement of physical QR codes at corresponding on-site locations, enabling precise monitoring of installation progress.

In Fig. 7, supervisors authenticate via a mobile app and scan QR codes between designated points (e.g., Point A to Point B) to retrieve installation data from the SQL Server. The system calculates quantity discrepancies in real time. For installations extending beyond threshold markers (e.g., Point C in Fig. 8), predefined allowances are applied. All captured data, tagged with relevant drawing codes, is transmitted to a cloud-based SQL database, where predefined queries cleanse records, flag unreliable entries (e.g., poor connectivity artefacts), and update the work breakdown structure progress.

Collectively, this closed-loop framework enhances progress monitoring accuracy, minimises manual errors, and elevates project transparency through BIM-intelligent data workflows.

3.3. Smartphone and database connection design

The initial stage entails the development of a user-friendly, HTML-based Graphical User Interface (GUI) to facilitate secure data entry via smartphones. Users are required to log in and input key project details, including login credentials, building identification, supervisor ID, installation quantities (e.g., cabling and cable tray), and the date of installation. This information is transmitted directly to a cloud-hosted SQL database. As further depicted in Fig. 9, the mobile application serves as the front-end platform for on-site data collection, while the back-end infrastructure manages data flow and integration. Through this application, supervisors can submit real-time updates on installed quantities (e.g., metres of cable tray and cables installed) along with any site-related issues or comments, all of which are securely recorded in the SQL Server database.

As illustrated in Fig. 3, generating real-time analytical and visual reports through the Analytical_DB requires a continuous and structured data flow. To enable this, the Project_DB functions as a centralised repository, consolidating two key data streams: (i) planned installation quantities and associated dates sourced from the Planning_DB, and (ii) actual on-site progress (e.g., installed metres) captured in real time via QR code scanning through the Site_Work interface. To facilitate the transfer of real-time data, a server-side PHP script is employed. This mechanism enables seamless communication between mobile data capture and back-end storage. Consequently, the Analytical_DB gains

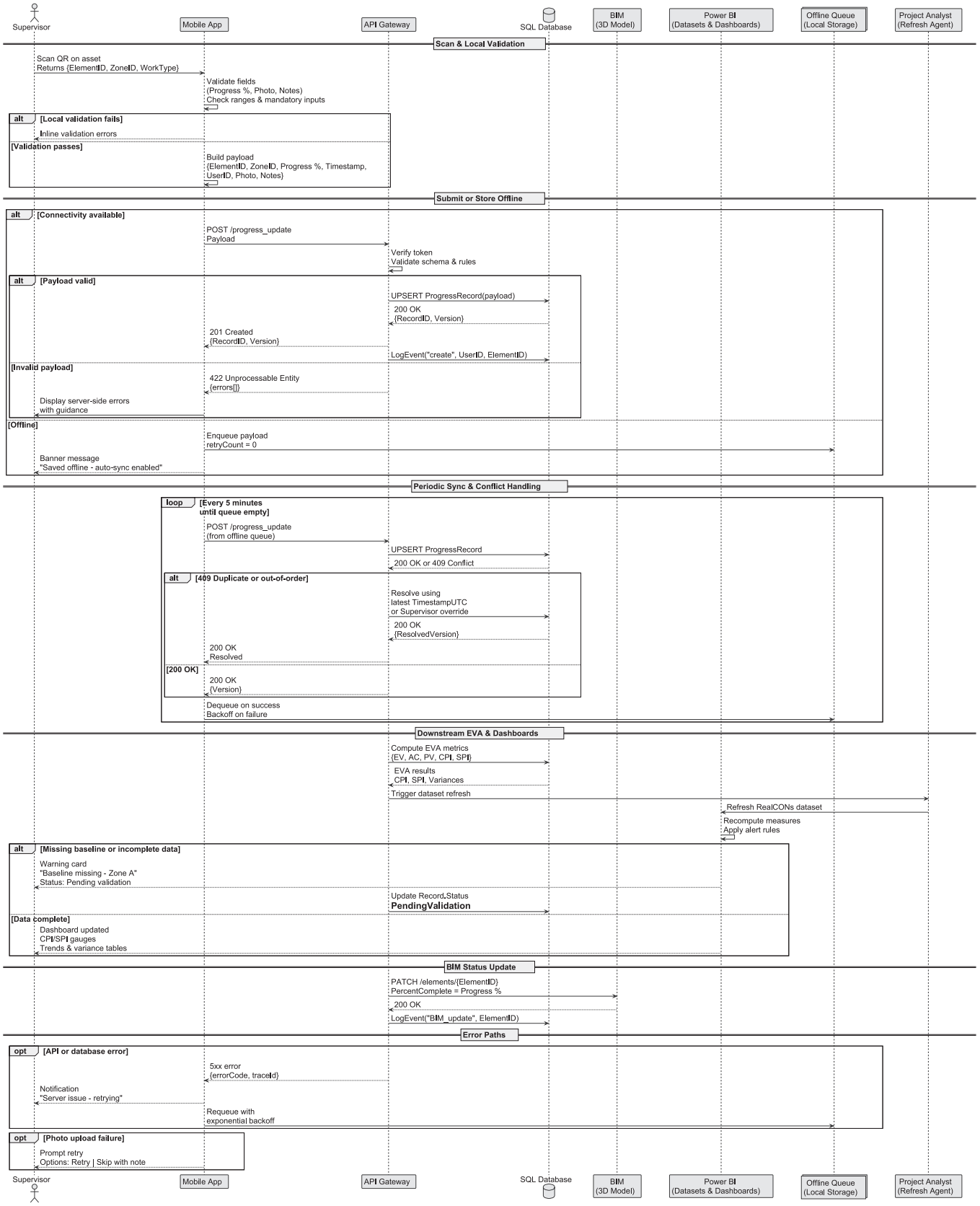


Fig. 4. Sequence diagram: Site to insight.

Table 4
Database tables and relationships.

Table Name	Primary Key	Connected Table	Connection Method	Purpose
BIM_Elements	Element_ID	QR_Code_Register	Foreign Key: Element_ID	Links BIM elements to QR codes
Planning_DB	Activity_ID	BIM_Elements	Attribute Match: WBS_Code	Activities associated with BIM elements
		Site_Work	Foreign Key: Activity_ID	Connects schedules to site records
Cost_Records	Cost_ID	Site_Work	Functional Link (unspecified FK)	Actual cost calculations

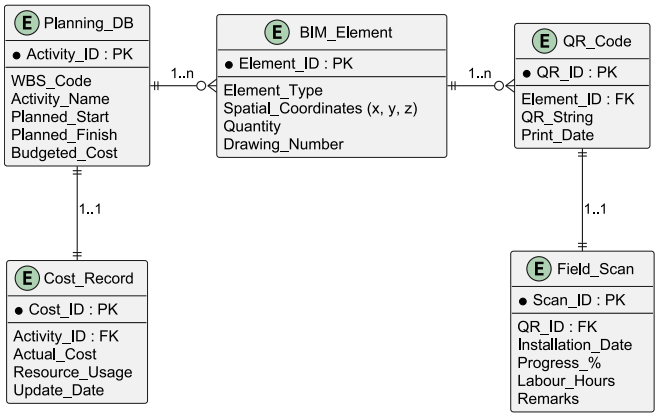


Fig. 5. Database schema and relationships.

Table 5
QR code data key fields (An example).

Field	Description	Example Value	Data Format
location	Physical installation site	"Block A, Level 3"	{"location": "Block A, Level 3", "drawing_number": "DRG-EI-2024-015", "building_code": "GHS*-EI-08", "quantity": 12, "unit": "m"}
drawing_number	Reference engineering drawing	"DRG-EI-2024-015"	
building_code	Project-specific asset ID	"GHS*-EI-08"	
quantity	Installed units (with unit type)	12 (meters, pieces)	
timestamp	Auto-generated scan time (ISO format)	2024–07-15 T14:30:00Z	

* *GHS* is a project name that has been introduced in section 3.5 – case study.

Table 6
QR Code generation.

Example Code	Sample
import qrcode, json data = {"location": "Block A, Level 3", "drawing_number": "DRG-EI-2024-015", "building_code": "GHS*-EI-08", "quantity": 12} img = qrcode.make(json.dumps(data)) img.save("construction_QR.png")	

access to a comprehensive dataset comprising both actual and planned metrics, as well as recorded delay event comments, thereby supporting robust performance tracking and informed decision-making.

3.4. Power BI design

As illustrated in Figs. 3 and 4, the RealCON framework's data workflow begins with the real-time on-site capture of progress updates and delay annotations via QR scanning, which are stored in a central SQL database (Project_DB). This repository integrates cost/schedule data from Planning_DB and BIM models, feeding Analytical_DB (Fig. 10), where custom DAX measures compute key Earned Value Management (EVM) metrics: Planned Value (PV), Earned Value (EV), and Actual Cost (AC), directly from linked BIM, schedule, and field data tables. The resulting visualisations enable early detection of delays and discrepancies through dynamic dashboards, providing real-time identification of cost/schedule deviations. This end-to-end pipeline enhances stakeholder communication and project oversight by supporting data-driven decisions across the construction lifecycle. So, a sample of Power BI DAX codes is as follows:

- Planned Value (PV): PV is derived from the budgeted cost of activities proportionate to planned progress by a given date. Dax is:

Planned Value (PV) =.
*CALCULATE(SUM('Planning_DB '(Budgeted_Cost)) *.*
DIVIDE(DATEDIFF('Planning_DB '(Planned_Start), TODAY(), DAY),
DATEDIFF ('Planning_DB '(Planned_Start), 'Planning_DB '(Planned_Finish), DAY)),
FILTER ('Planning_DB ', TODAY() >= 'Planning_DB '(Planned_Start) && TODAY() <= 'Planning_DB '(Planned_Finish)))

- Earned Value (EV): EV links actual field scans of BIM elements to scheduled activities via the WBS. Dax is:

*Earned Value (EV) = SUMX('Field_Scan', RELATED('Planning_DB '(Budgeted_Cost)) *.*
('Field_Scan'(Progress_%) / 100))

- Actual Cost (AC): AC is drawn directly from cost records linked to each scheduled activity. Dax is:

Actual Cost (AC) = SUM('Cost_Record'(Actual_Cost)).

These measures are embedded in Power BI dashboards, enabling stakeholders to compare PV, EV, and AC in real-time. This supports proactive identification of variances (Cost Variance, Schedule Variance) and forecasting metrics (CPI, SPI). By exposing the underlying DAX code, this paper ensures complete transparency and reproducibility of the analytical workflow, addressing the reviewer's request for technical detail.

3.5. Case study

The Electrical and Instrumentation (E&I) trade was selected for testing the RealCON framework's field applicability, as it represents a critical service trade with distinctive sequencing requirements. E&I works (comprising cable containment installation, cable pulling, equipment installation, termination, testing and commissioning) must precede dependent trades such as mechanical and data/security systems, which require power supply for their commissioning. While limited task overlap may be permissible, this precedence relationship makes E&I activities particularly suitable for validating real-time progress monitoring.

3.5.1. Case study introduction and selection

All case studies focus on two-storey, high-security buildings, each

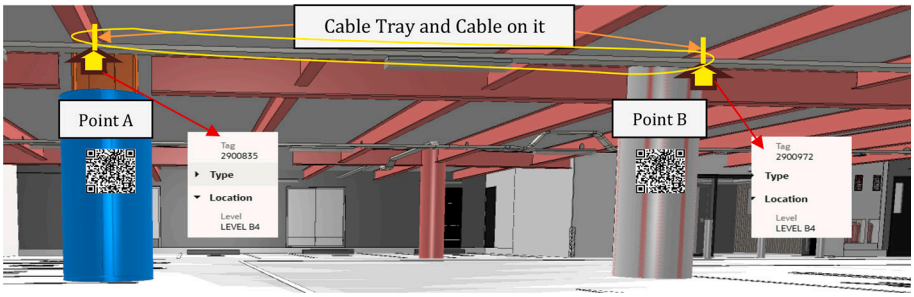


Fig. 6. QR codes display locations.



Fig. 7. Point to point.



Fig. 8. Point to threshold.

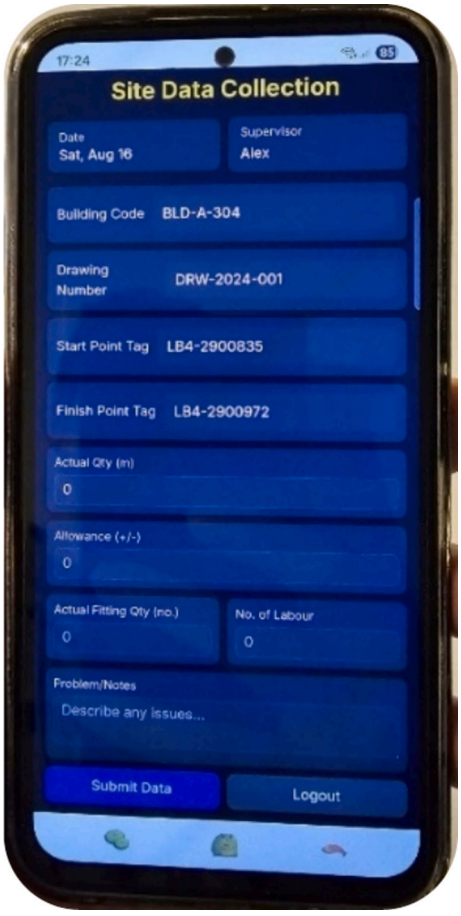


Fig. 9. GUI on a smartphone.

with a floor plan of roughly 900 m^2. Given the stringent security requirements, these projects demand meticulous monitoring of time, cost, and quality throughout their life cycle. Two critical factors underscore this need: (1) the high cost of materials and labour, and (2) the necessity of strict timeline oversight. Consequently, daily productivity tracking is crucial for maintaining efficiency and adhering to schedules. However, stringent security protocols prohibit the use of RFID tags, embedded sensors, and aerial drones, significantly constraining data collection and monitoring methods.

The case studies were chosen based on specific criteria to ensure they align with the challenges addressed by the RealCONs framework: 1) Scale and Value – The selected projects, valued between $90 M and $100 M, represent mid-to-large-scale developments, offering a broad perspective on project complexities; 2) Electrical Scope – Variations in cabling (10,890–14,000 m), containment (12,000–15,800 m), and fittings (120–150 units) provide a robust dataset for analysing delay factors across different work scopes; 3) Geographic Similarity – All projects were executed under comparable contract structures and construction environments, facilitating consistent evaluation of the RealCONs framework; 4) Data Availability – Each project generated multiple status reports including progress reports (manual forms, MS Excel, MS Project/Primavera) and delay events reports (MS Excel). However, reliance on disparate data sources has compromised accuracy, delayed reporting, and increased response times in both internal (e.g., site supervisors to project managers) and external (e.g., subcontractors to design teams) interactions; 5) Project Stage Similarity – Roofing work is complete across all projects, granting E&I teams site access. Construction progress exceeds 85%, and each project is managed by two executive teams of 15 members, with identical reporting mechanisms; and 6) Stakeholders such as project managers, planners, quantity surveyors, supervisors, and engineers were involved.

Table 7 illustrates that all four selected projects are comparable in terms of scale and value.

3.5.2. *Implementation of the REALCONS framework*

The implementation of the RealCONs framework in an active construction environment follows a structured sequence, ensuring that data from on-site activities is accurately captured, processed, and transformed into actionable insights. For the Electrical and Instrumentation (E&I) trade, this process is illustrated using the example of cable containment installation and subsequent cable pulling. For these

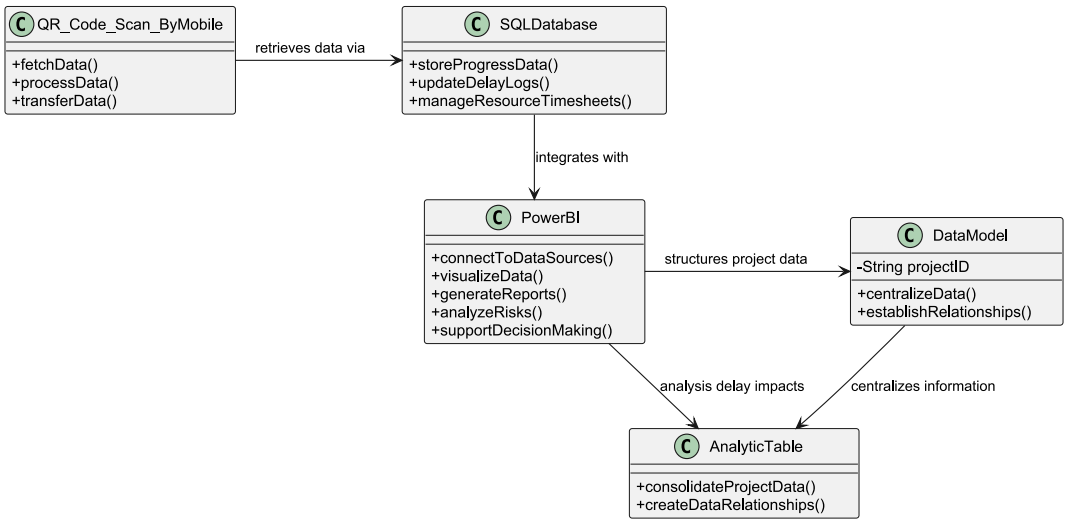


Fig. 10. Power BI overview model.

Table 7
Selected projects details: E&I scope.

Project E&I Specs	GHS	KIT
Value (million $)	100	100
Cabling (meter)	12,000	11,890
Cable Containment (meter)	13,800	13,000
Fitting (No)	135	138

purposes, five steps are designed in Fig. 12: 1) Preparation and System Setup, 2) Data Capture at the Project Entry Gate, 3) Data Transmission and Storage, 4) Data Processing and Analytics, 5) Visualisation and Reporting in Power BI.

From Fig. 12, the BIM model is configured to define spatial locations, building codes, and drawing references for E&I works such as tray installation and cabling. Each section is assigned unique tags to ensure precise traceability, and QR codes are generated using a Python script to encode attributes including location identifiers, tray lengths, and cable quantities. A secure SQL Server (Project_DB) is established in parallel (see section 3), structured into fields such as installation dates, material costs, and labour. To support on-site data capture, a mobile application (Graphical User Interface - GUI) has been developed using HTML and Python, while PHP scripts connect the interface to the SQL Server. API permissions are formally obtained from the BIM (Revit) and Planning (Primavera P6) teams, enabling the server to draw element-specific data directly from design and planning systems. On-site, QR codes are printed and fixed at BIM-defined locations. Supervisors log into the mobile application, scan the codes, and automatically retrieve the associated building code, drawing number, and start-point tag stored in the SQL Server. Progress is recorded by scanning the corresponding finish-point tag. Where the measured length matches the BIM-defined element, the record is confirmed; where discrepancies occur, a pre-agreed tolerance (e.g., ±10%) is applied. These verified records are transmitted securely to the SQL Server and Power BI, ensuring an accurate, real-time record of progress while accommodating permissible site variations.

The RealCON Power BI model integrates three core data streams for comprehensive project monitoring: 1) Planned Data (baselines and forecasts from Planning_DB); 2) Actual Site Data (real-time progress captured via QR scanning and stored in SQL Server); and 3) Calculated Metrics (analytical outputs comparing planned vs. actual values). This integration enables automated Earned Value analysis (Table 8), with data aligned to WBS/CBS structures. Key metrics, such as SPI and CPI, are calculated in real-time, flagging deviations (e.g., cable installation delays despite tray completion) for immediate intervention. The framework automatically detects variances, for instance, when only 30 m of cable tray or cabling is installed against a 50 m target, and triggers alerts for resource reallocation or task resequencing.

Table 8
Earned value metrics [81].

Parameters	Formula	Warning Indexes
Schedule Variance (SV): Deviation between earned progress and planned progress (time-based)	SV = EV - PV	SV < 0: Behind schedule (PV > EV)
Cost Variance (CV): Deviation between earned value and actual cost (cost-based)	CV = EV - AC	CV < 0: Over budget (AC > EV)
Cost Performance Index (CPI): Efficiency of cost utilisation (value earned per $ spent)	CPI = EV / AC	CPI =1: on budget CPI >1: Under budget CPI <1: Over budget
Schedule Performance Index (SPI): Efficiency of time utilisation (progress vs. plan)	SPI = EV / PV	SPI =1: on schedule SPI >1: ahead of schedule SPI <1: Behind schedule
Estimate At Completion (EAC): Forecasted total project cost based on current performance	EAC=BAC/ CPI	EAC > BAC: Likely cost overrun

Planned Value (PV): Budgeted cost of work scheduled up to a given date (BCWS): How much work should be done?
Actual Cost (AC): Actual cost incurred for work performed up to a given date (ACWP): How much did the work cost?
Earned Value (EV): Budgeted cost of work actually completed up to a given date (BCWP): How much work was done?
Budget at Completion (BAC) is the sum of all budgets allocated to a project scope.

As the analytical layer, Power BI transforms validated SQL data using DAX coding into interactive dashboards featuring: i) Progress curves and productivity rates; ii) BIM-linked work zone status maps; iii) EV trend charts (CPI/SPI); and iv) Budget vs. Actual cost tracking. The cross-device accessible reports enhance transparency in decision-making, improving delay management and accountability across E&I works (see Fig. 11 and Table 9 for the model architecture and constituent tables).

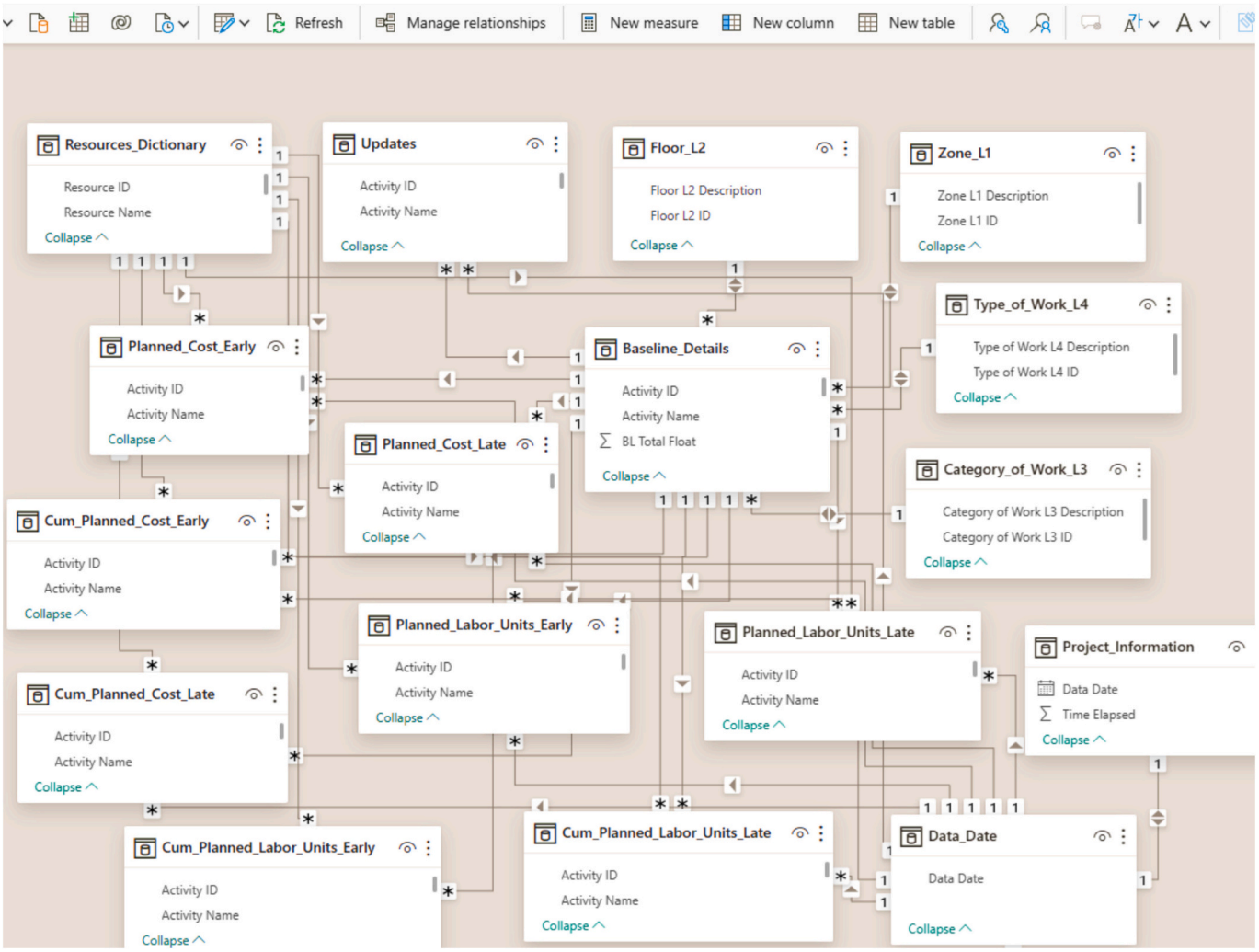


Fig. 11. Data model.

Table 9
Power BI data tables grouped by source type.

Category	Tables / Outputs		Purpose
Planned Data	- Zones and Work Types - Resources_Dictionary - Baseline_Details	- Planned Cost and Labour - Cumulative Plans - Project_Information	Defines project structure (zone/floor/work type), resource catalogues, baseline labour and cost data, and aggregated forecasts.
Actual Site Data	- Updates - Data_Date		Captures site progress (dates, status, quantities) and logs reporting dates for version control.
Calculated Metrics	- Planned vs Actual - Forecasting	- Delay Analysis - Zone/Floor Analysis	Compares planned vs. actual data, forecasts resource/cost needs, identifies schedule delays, and enables location-specific performance review.

4. Test and validation of the proposed framework (REALCONS)

The proposed RealCONs framework was tested and validated using real-time data from two major construction projects involving electrical and instrumentation (E&I) works. The objective was to evaluate its effectiveness in real-time project tracking, with emphasis on early delay identification and management. The analysis is structured into two parts: (1) Data Continuity and (2) Statistical Analysis. RealCONs' performance (S1) was compared with an existing reporting system (S2) over a 90-day data collection period, which aligned with the contractual duration of Construction Phase 1 (Table 10).

4.1. Data continuity

Daily earned value (EV) data were used to assess reporting continuity (Table 10). Days with EV = $0.00 were treated as missing and excluded from performance calculations, although such gaps did not prevent invoice payments (based on Actual Cost, AC). For S1 (GHS), only five days had missing EV data (Days 21, 28, 39, 74, and 86). In contrast, S2 (KIT) recorded 35 missing days (e.g., Days 2–3, 11–14, 24–26, etc.), indicating significantly poorer data continuity (Fig. 13; Table 11). Excluding duplicates (e.g., day 74, missing in both projects), the dataset contained 39 unique missing days ($40 - 1 = 39$). This resulted in 51 ($90 - 39 = 51$) paired daily observations (n) for statistical comparison.

A proportion test for missing data (Table 11) revealed a significant disparity in reporting quality, with RealCON (S1) exhibiting nearly

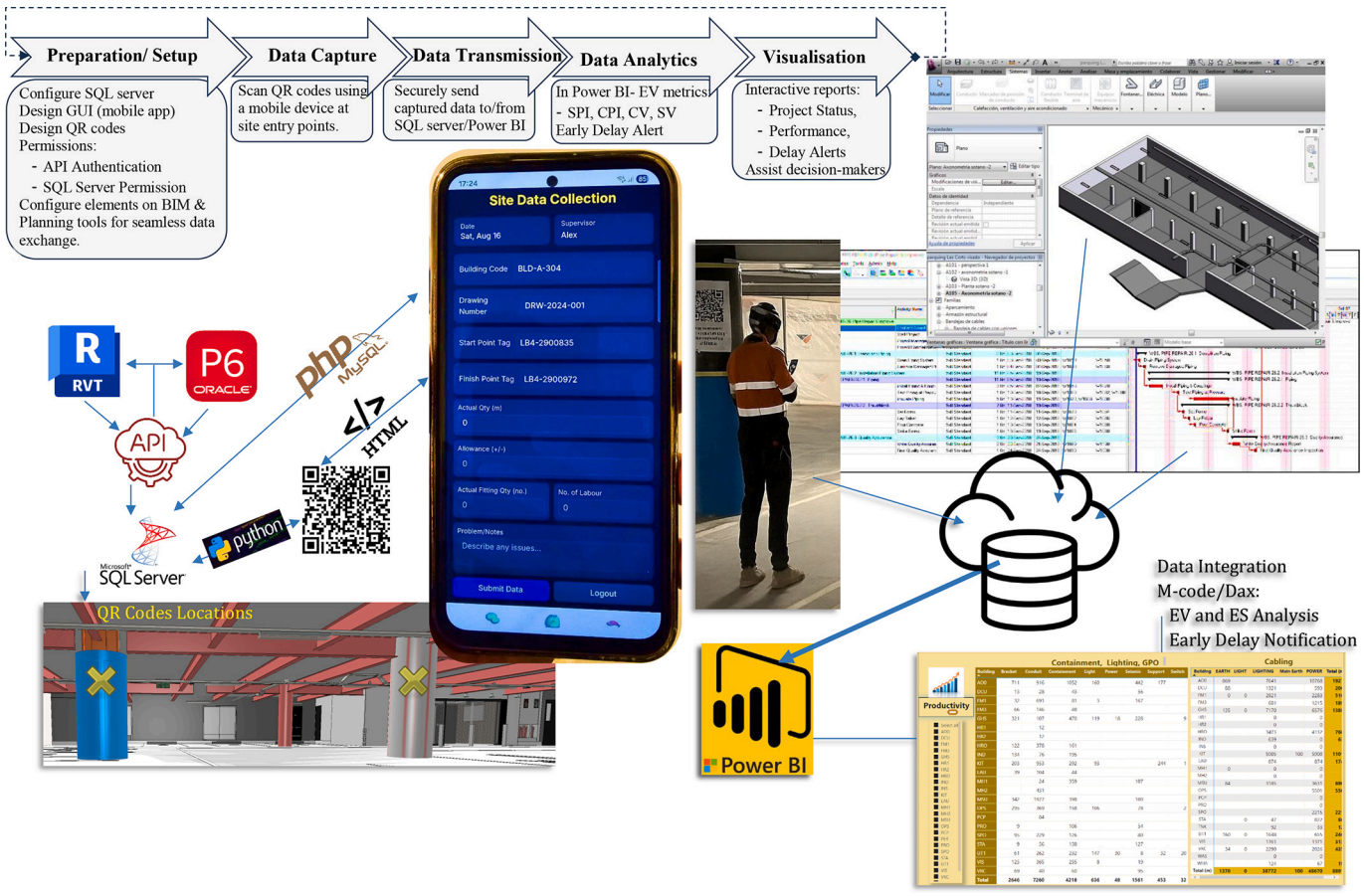


Fig. 12. RealCONs implementation steps.

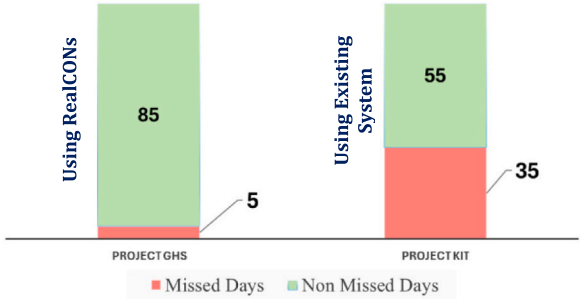


Fig. 13. Daily received reports.

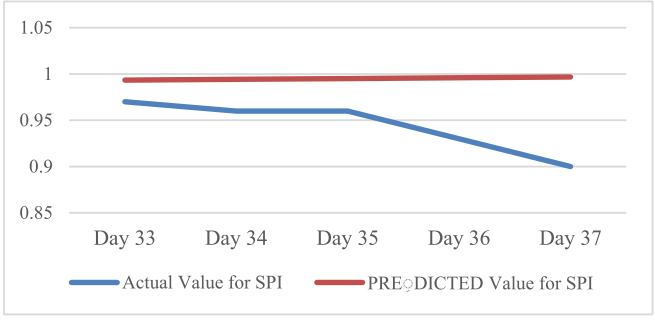


Fig. 14. SPI trend over five days.

seven times fewer missing data points than Site 2 (S2). A two-proportion test confirmed this difference was statistically significant ($\chi^2 = 28.929$, df = 1, *p* < 0.001). The effect size, calculated using the Phi coefficient, was $\varphi = 0.401$, which is considered to approach a significant effect based on Cohen's criteria. The finding that RealCONs (S1) substantially improves data continuity is supported by both a significant chi-square test and a confidence interval (CI) for the difference in proportions of days with missing data. This interval, which excludes zero, confirms the statistical significance of the difference at the 95% confidence level: P1 = 5/90 = 0.0556 and P2 = 35/90 = 0.3889, then CI = (0.3333 ± 1.96) × 0.059 = (0.218,0.449) while 0.059 is the Standard Error and 0.3333 is P2 − P1. Moreover, Section 4.2 addresses the data normality test, descriptive statistical analysis, the Mann–Whitney *U* test for CPI and SPI, the Wilcoxon signed-rank test, regression analysis of SPI, Earned Schedule (ES) analysis, and cost and schedule forecasting (EAC and TEAC).

4.2. Statistical analysis

4.2.1. Tests of normality

To evaluate the data and select an appropriate statistical test (parametric or non-parametric), the distribution of the paper variables was assessed for normality using the Shapiro-Wilk test (see the Tests of Normality table). The results indicated that, at the 95% confidence level, the assumption of normality was not confirmed for the variables in both projects (S1 and S2). Consequently, the use of parametric tests was not permissible, and the non-parametric Mann-Whitney U and Wilcoxon Signed-Rank tests were employed instead.

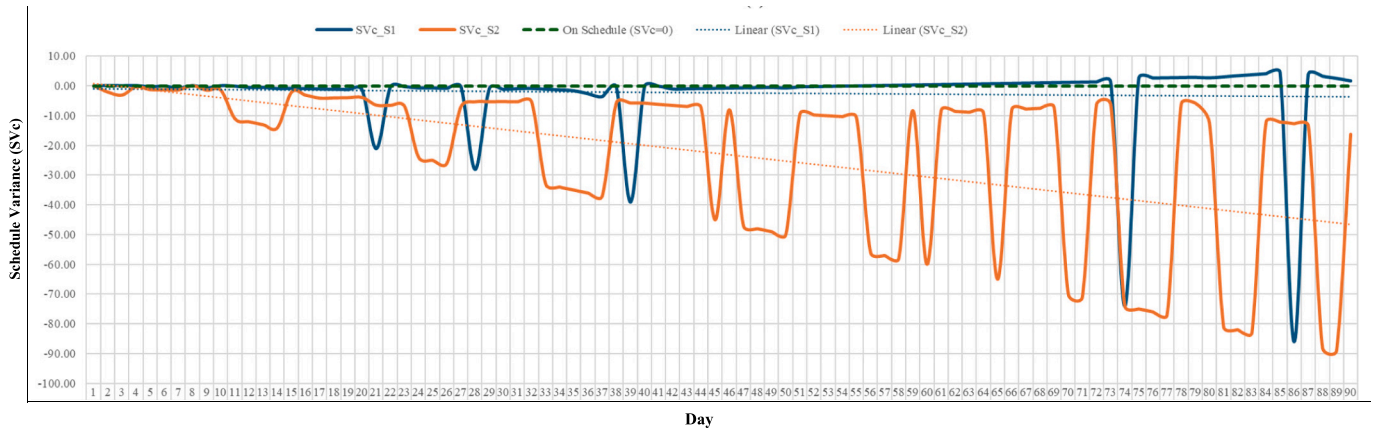


Fig. 15. Schedule Variance (SVc).

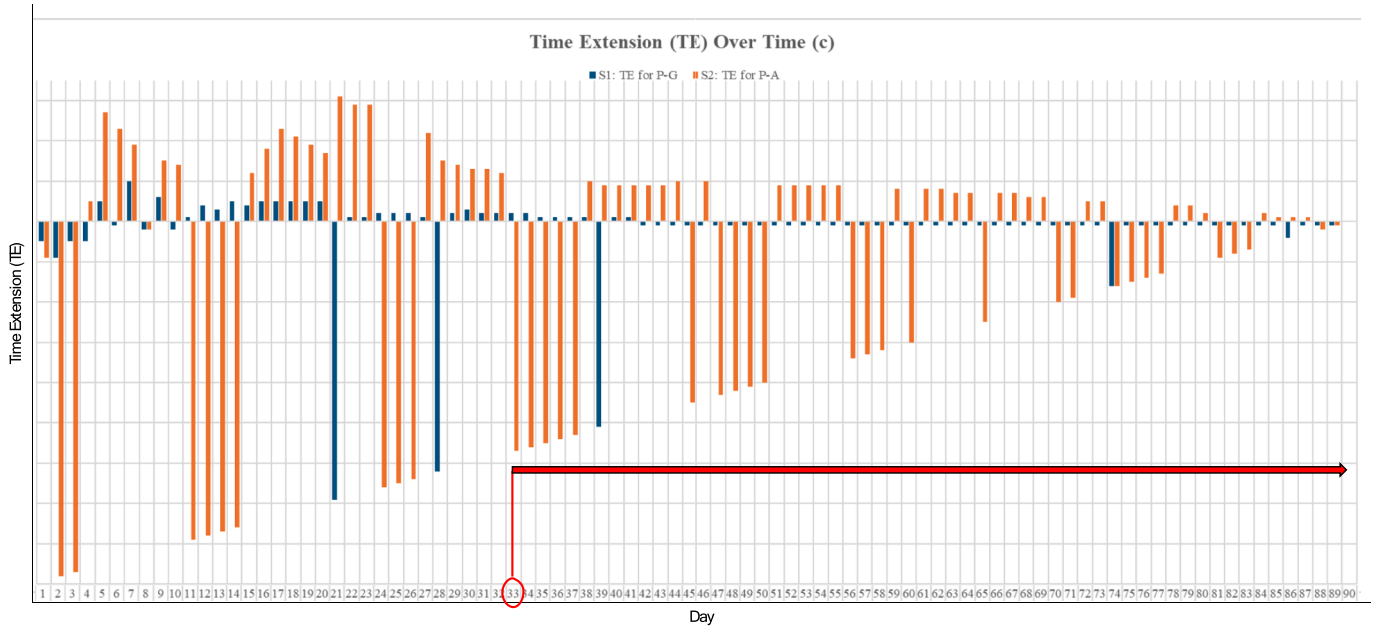


Fig. 16. Time extension: RealCONs (S1) Vs existing (S2)

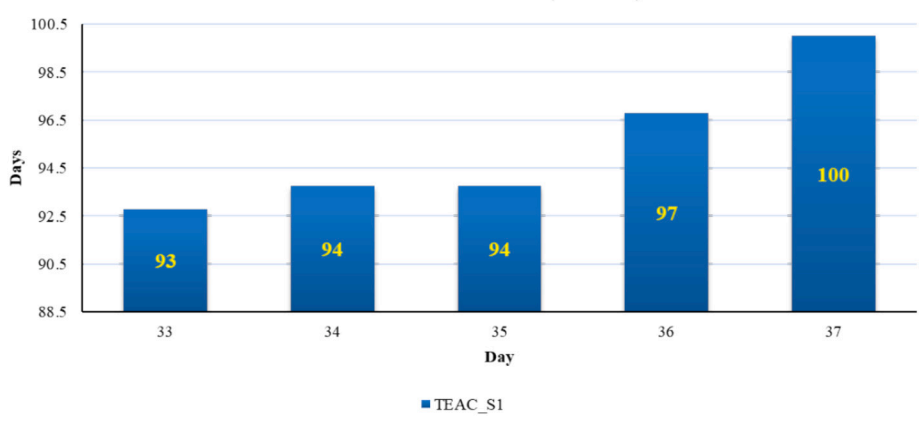


Fig. 17. Forecasted Duration (TEAC).

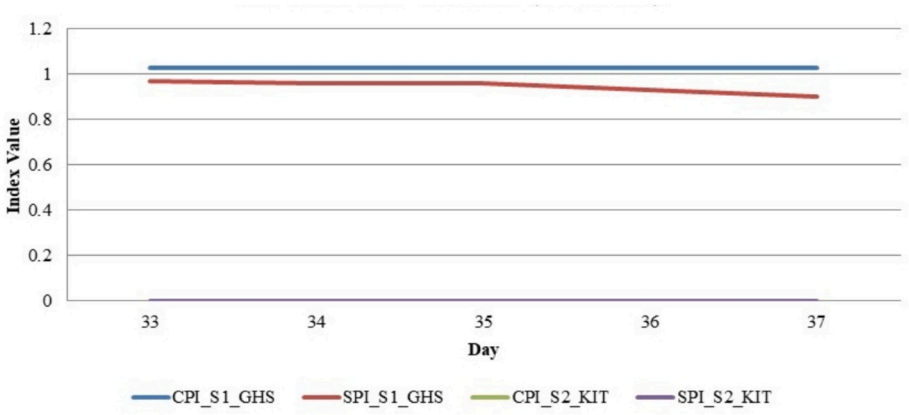


Fig. 18. CPI and SPI Trends (S1 vs S2)

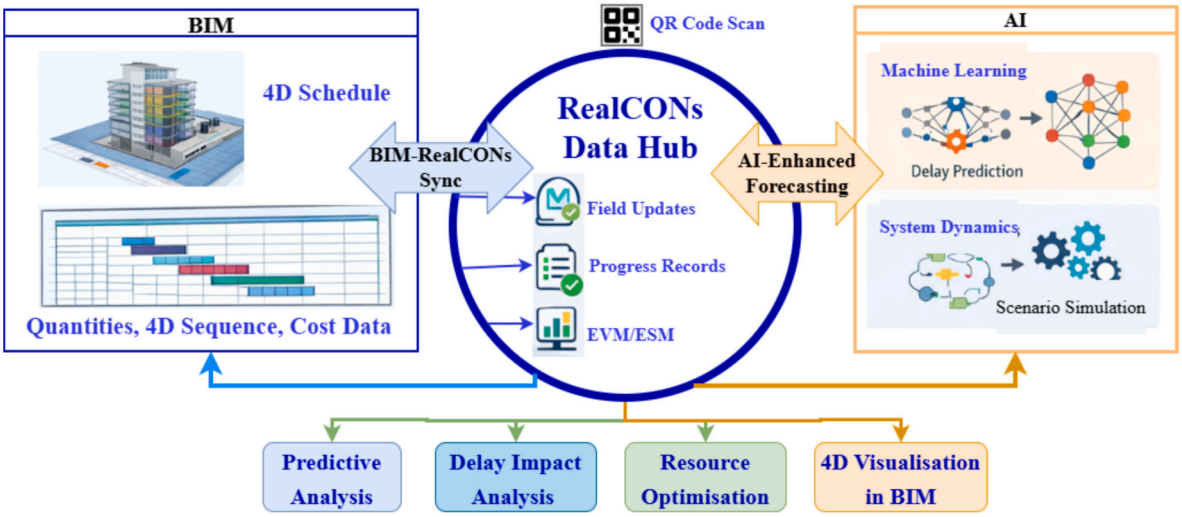


Fig. 19. Future integration of BIM and AI with RealCONs.

Table 11
Proportion test for missing data.

			Status			Total
			Missed Day	Normal Day		
Project	RealCONs (GHS-S1)	Count	5	85		90
		% within Project	5.6%	94.4%		100.0%
	Exist (KIT-S2)	Count	35	55		90
		% within Project	38.9%	61.1%		100.0%
	Total	**Count**	**40**	**140**		**180**
		% within Project	**22.2%**	**77.8%**		**100.0%**
	Pearson Chi-Square				28.929[a]	
	df				1	
	Asymptotic Significance (2-sided)				0.000	

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 20.00.

Table 12
Descriptive statistics for metrics.

Project	Metrics	N	Mean	Median	Std. Deviation	Range
Both	**PV**	90	$50,555.56	$50,555.56	$29,027.45	$98,888.89
GHS	**GHS_AC**	90	$58,405.69	$45,995.70	$41,681.39	$138,218.80
	GHS_CPI	85	$0.96	$1.02	$0.13	$0.40
	GHS_CV	90	-$10,129.67	-$138.29	$21,427.76	$136,690.00
	GHS_EV	85	$51,115.79	$51,500.00	$30,532.31	$100,833.33
	GHS_SPI	85	$1.00	$1.02	$0.04	$0.20
	GHS_SV	90	-$2279.53	$225.11	$14,665.44	$100,611.12
KIT	**KIT_AC**	90	$82,522.80	$76,277.30	$58,739.38	$198,322.50
	KIT_CPI	55	$0.58	$0.55	$0.12	$0.49
	KIT_CV	90	-$58,514.03	-$44,559.80	$52,169.11	$196,288.05
	KIT_EV	55	$39,287.09	$39,450.00	$23,916.65	$80,777.78
	KIT_SPI	55	$0.82	$0.81	$0.06	$0.40
	KIT_SV	90	-$26,546.78	-$13,888.89	$28,800.25	$99,000.50

Tests of Normality Table.

	Shapiro-Wilk		
	Statistic	Df	Sig.
KIT_EV	0.951	51	0.033
KIT_AC	0.927	51	0.004
KIT_CV	0.878	51	0.000
KIT_SV	0.951	51	0.034
KIT_CPI	0.935	51	0.008
KIT_SPI	0.789	51	0.000
GHS_EV	0.945	51	0.020
GHS_AC	0.919	51	0.002
GHS_CV	0.776	51	0.000
GHS_SV	0.929	51	0.004
GHS_CPI	0.912	51	0.001
GHS_SPI	0.903	51	0.001

4.2.2. Descriptive statistics

Before formal statistical tests, a descriptive analysis of the core performance metrics: the Cost Performance Index (CPI) and Schedule Performance Index (SPI), was conducted to characterise the fundamental behaviour of each project. These indices were calculated daily for both case studies using the standard formulae (CPI = EV/AC; SPI = EV/PV), as detailed in Table 8. To ensure analytical integrity and avoid distortion from non-productive periods, calendar days with an Earned Value (EV) of zero were excluded from the dataset. This yielded 85 valid daily observations for the GHS project (Site 1) and 55 for the KIT project (Site 2). The resultant statistics, summarised in Table 12, reveal a stark dichotomy in project performance. For Site 1 (GHS), both the CPI and SPI distributions exhibit means closely clustered around 1.00, with comparatively low standard deviations. This indicates that the project was consistently executed close to its budgeted cost and planned schedule, with minimal volatility.

In contrast, the data for Site 2 (KIT) presents a consistently challenged performance profile. The mean values for both indices are substantially below the nominal benchmark of 1.00, confirming systematic cost and schedule overruns. Furthermore, the significantly higher measures of dispersion (e.g., standard deviation, range) point to pronounced instability and high variability in its weekly performance. This descriptive delineation not only validates the initial project selection as comparative cases but also provides essential context for the inferential analyses that follow, suggesting that the underlying causes of performance variation warrant rigorous investigation.

4.2.3. Impact of missing data

For the comparison across the missing data role in projects S1 (GHS) and S2 (KIT), a between-group pairwise comparison was conducted for each variable (CPI, SPI and others), using the Mann-Whitney *U* test for nonnormal variables (Section 4.2.1). In Table 13, the data across groups were treated as independent because each project held different

Table 13
Mann-Whitney *U* test on CPI and SPI (After missing vs. all other days).

Project	Variable	Group	N	Mean Rank	Mann-Whitney U	Wilcoxon W	Z	Asymp. Sig. (2-tailed)
GHS (S1)	**EV**	After Missing	5	3.00	0.000	15.000	−3.743	0.000
		All Other	85	48.00				
	AC	After Missing	5	49.60	192.000	3847.000	−0.361	0.718
		All Other	85	45.26				
	CV	After Missing	5	8.40	27.000	42.000	−3.268	0.001
		All Other	85	47.68				
	SV	After Missing	5	3.00	0.000	15.000	−3.743	0.000
		All Other	85	48.00				
	CPI	After Missing	5	3.00	0.000	15.000	−3.750	0.000
		All Other	85	48.00				
	SPI	After Missing	5	3.00	0.000	15.000	−3.762	0.000
		All Other	85	48.00				
KIT (S2)	**EV**	After Missing	35	18.00	0.000	630.000	−8.211	0.000
		All Other	55	63.00				
	AC	After Missing	35	48.74	849.000	2389.000	−0.939	0.348
		All Other	55	43.44				
	CV	After Missing	35	31.40	469.000	1099.000	−4.085	0.000
		All Other	55	54.47				
	SV	After Missing	35	22.23	148.000	778.000	−6.741	0.000
		All Other	55	60.31				
	CPI	After Missing	35	18.00	0.000	630.000	−8.214	0.000
		All Other	55	63.00				
	SPI	After Missing	35	18.00	0.000	630.000	−8.221	0.000
		All Other	55	63.00				

Table 14
Wilcoxon signed-rank test on 90 paired days.

Index	Median (S1)	Median (S2)	Negative Ranks*	Positive Ranks**	Ties***	Z-score	*p*-value
EV	$51,500.0000	$39,450.0000	(*N* = 50,Mean = 26.5)	(*N* = 1,Mean = 1)	0	−6.205	0.000
AC	$45,995.7000	$76,277.2950	(*N* = 0,Mean = 0)	(*N* = 90,Mean = 45.5)	0	−8.239	0.000
CV	-$138.2900	-$44,559.7950	(*N* = 86,Mean = 46.63)	(N = 4,Mean = 21.25)	0	−7.896	0.000
SV	$225.1050	-$13,888.8900	(*N* = 84,Mean = 44.71)	(N = 5,Mean = 49.9)	1	−7.172	0.000
CPI	1.0200	0.5500	(*N* = 51,Mean = 26)	(N = 0,Mean = 0)	0	−6.218	0.000
SPI	1.0200	0.8100	(N = 50,Mean = 26.48)	(N = 1,Mean = 2)	0	−6.202	0.000

* Negative (e.g. KIT<GHS), ** Positive (e.g. KIT>GHS), ***Tie (e.g. KIT = GHS)

collected data.

From Table 13, both projects exhibited a statistically significant collapse in cost and schedule efficiency during the "After Missing" periods (U = 0.000, *p* < 0.001 for all CPI/SPI tests). However, the larger mean rank disparity in the KIT project ("After Missing" Mean Rank = 18.00 vs. "All Other" = 63.00) compared to GHS ("After Missing" Mean Rank = 3.00 vs. "All Other" = 48.00) suggests that while the performance drop is absolute in both cases, its impact on the larger KIT project portfolio is more pronounced. Furthermore, the issue in KIT stems from a substantially higher number of missing data events (*N* = 35 vs. *N* = 5 in GHS), indicating a more systemic data management issue.

4.2.4. Daily performance comparison (51 paired days)

For GHS (S1), the two subsequent tables are the findings from a linear regression analysis modelling the SPI against the independent variable, Day. The Analysis of Variance (ANOVA) results, detailed in the first table, demonstrate that the overall regression model is statistically significant (F-test *p*-value = .000 < α = 0.05). Furthermore, the *t*-test results, contained within the second table, also report a significance value of 0.000 ($p < .05$). This provides sufficient evidence to reject the null hypothesis that the coefficient for the regression slope is zero. Consequently, the estimated linear regression model is given by the following equation: SPI = 0.965 + 0.575xDay.

ANOVA (GHS).

Model		Sum of Squares	df	Mean Square	F	Sig.
GHS	Regression	0.041	1	0.041	41.020	.000[b]
	Residual	0.084	83	0.001		
	Total	0.125	84			

b. Predictors: (Constant), Day

Coefficients (GHS).

Model	Unstandardised Coefficients		Standardised Coefficients	t	Sig.	Confidence Interval (CI) for B	
	B	Std. Error	Beta			Lower Bound	Upper Bound
(Constant)	0.965	0.007		139.532	0.000	0.952	0.979
Day	0.001	0.000	0.575	6.405	0.000	0.001	0.001
R Square				0.331			

The analysis aimed to check whether there were any significant differences in project performance on different days. A within-group pairwise comparison was conducted for the S1 and S2 projects using the Wilcoxon signed-rank test for non-normal variables. In Table 14, a direct comparison was made on the 90 days (n), showing that both results (for CPI and SPI) confirm significantly stronger performance in S1, as follows:

4.2.5. Analysis of behaviour during a delay event (days 33–37)

From Table 10, a significant delay occurred between Days 33 and 37. During this period, S1 captured detailed performance trends, whereas S2 reported no earned value (EV = $0), rendering EVM-based forecasting impossible. This section is divided into I) Regression analysis of SPI (S1 and S2); II) Earned Schedule (ES) Analysis, III) Cost and schedule Forecasting (EAC and TEAC); IV) CPI and SPI Behaviour (Days 33–37); and V) Managerial implications.

I. **Regression analysis of SPI (S1 and S2)**

The analysis indicates that the model demonstrates strong predictive validity for the SPI under the tested conditions, despite a five-day delay event. Fig. 14 shows a generally robust performance, with the model accurately replicating the observed trend and variability. The close fit between predicted and observed values suggests an absence of pervasive systematic bias, as errors are localised rather than widespread. However, a specific calibration need is identified. While a near-perfect alignment was achieved on Day 34, a consistent overprediction from Days 35 to 37 reveals a temporary positive bias. This pattern confirms the model's fundamental utility while pinpointing a defined scope for refinement to enhance future forecast precision.

In contrast, for KIT (S2), the ANOVA results indicate that the regression model is not statistically significant, with an F-test significance value of 0.229 ($p > .05$). Furthermore, the *t*-test results for the regression slope coefficient show a non-significant relationship between Day and SPI ($p = .229$). As this exceeds the 0.05 alpha level, the null hypothesis, that the regression slope is equal to zero, cannot be rejected.

ANOVA (KIT).

Table 15
Earned Schedule Management (ESM) matrix.

	Planned Value	Earned Value		Earned Schedule		Increment (I)		Schedule Variance		Schedule Performance	
Day (C)	**PV-S1 & S2**	**EV-S1**	**EV-S2**	**ES-S1**	**ES-S2**	**I-S1**	**I-S2**	**SVc-S1**	**SVc-S2**	**SPIc-S1**	**SPIc-S2**
1 …	1166.67	1222.2	1.05	1.1	0.05	0.1	0.05	0.1	1.05	1.1	1166.67
… 90	100,000	102,000	82,000	91.8	73.8	1.8	−16.2	1.8	−16.2	1.02	0.82
Column "I" denotes the increment index, and "C" indicates the actual time (day)											
ESM [81]	$ES=C+I$			C: is the number of periods completed for BCWP≥ BCWS							
	$I=(EV\text{-}PV_C)/(PV_{C+1}-PV_C)$			I is an increment index and uses PV and EV from EVM							
	$SV_C=ES-C$			Earned Schedule Variance (SV_C): Earned Schedule – Actual Time							
	$SPI_C=ES/C$			Earned Schedule Performance Index SPI_C: Earned Schedule/Actual Time							

From Table 16 (a cut view of Table 15), during days 33–37, both projects exhibit concerning schedule performance, but with differing levels of severity.

Model		Sum of Squares	df	Mean Square	F	Sig.
KIT	Regression	0.006	1	0.006	1.479	.229[b]
	Residual	0.208	53	0.004		
	Total	0.214	54			

b. Predictors: (Constant), Day

Coefficients (KIT).

Model	Unstandardised Coefficients		Standardised Coefficients	t	Sig.	Confidence Interval (CI) for B	
	B	Std. Error	Beta			Lower Bound	Upper Bound
(Constant)	0.838	0.016		50.833	0.000	0.805	0.871
Day	0.000	0.000	−0.165	−1.216	0.229	−0.001	0.000
R Square				0.027			

II. Earned Schedule (ES) Analysis

Earned Schedule Management (ES) extends traditional Earned Value (EV) by converting cost-based performance metrics into time-based indicators, allowing for a more reliable assessment of schedule performance. Unlike SPI and Schedule Variance (SV), which lose diagnostic power once a project passes its planned completion point, ES maintains interpretability throughout the project lifecycle [82]. By expressing progress in units of time rather than cost, ES enables more accurate detection of schedule slippage, supports early warning signals, and improves forecasting of completion dates. This strengthens decision-making for delay mitigation, resource adjustment, and performance benchmarking. ES has become increasingly important in complex or long-duration projects where conventional EVM metrics typically mask latent schedule risks [82]. This finding supports evidence that Earned Schedule (ES) outperforms traditional Earned Value Management for early detection of schedule deviations. Table 15 presents an Earned Schedule Management analysis of a specific delay event (Days 33-37) within the project's first 90 days, showing key metrics such as Earned Schedule (ES), Schedule Variance (SV(c)), and SPI(c). The complete data is in Appendix 2.

Project S2 shows a complete work stoppage, with EV remaining at zero and ES fixed at 0, accompanied by critically deteriorated metrics (SVc falling from –33 to –37 and SPIc remaining at 0). These late-stage disruptions mirror earlier zero-EV periods on days 11–14 and 24–26, where SVc had already dropped to –11 by day 11, indicating that continuous Earned Schedule monitoring would have signalled actionable delays from the outset. In contrast, Project S1 does not experience total cessation; however, EV stagnates from days 35–37, causing ES to plateau at 33.456 and resulting in steadily worsening schedule indicators (I and SVc shifting from –1.023 to –3.544; SPIc declining from 0.969 to 0.904). Earlier warning signs were also evident for S1, including a zero-EV day on day 21 (ES=0; SVc=–21; SPIc=0), suggesting emerging inefficiencies long before the later stagnation. While S1's deterioration is moderate compared with S2's critical halt, both projects display escalating schedule risk. Accordingly, Project S2 requires an immediate recovery strategy to address the causes of its repeated work stoppages, whereas Project S1 needs prompt intervention to halt further decline. For both projects, implementing structured weekly ESM monitoring and using trend patterns to forecast delays are essential proactive measures for preventing escalation into severe schedule disruption. However, Figure 15's daily SVc trends show S1 beginning slightly ahead (SVc $>$ 0) before stabilising near zero, indicating a consistent pace with minor variations. Conversely, S2 develops increasingly hostile SVc after Day 33, reflecting growing schedule slippage. The comparative analysis demonstrates how early efficiency losses in S2 compound into substantially larger delays than S1's more gradual slowdown.

SPIc and ES are critical for quantifying the impact of the Day 33 delay on project duration. These metrics evaluate the impact of cost and schedule efficiency on the remaining work. SPI metric, presented in Table 17 using the Time Extension (TS) formula, provides one approach for calculating additional days required:

From Table 17, a comparison of schedule delay estimation methods illustrates markedly different risk profiles for the two projects. For Project S1, both indicators show only minor slippage on day 33, with the SPI-based forecast suggesting an additional 1.82 days of delay and the Earned Schedule measure indicating 1.023 days already lost. This divergence reflects their conceptual distinction: the SPI method extrapolates current efficiency into future performance, whereas the ES method quantifies delay already incurred. In contrast, Project S2 exhibits critical schedule distress. The SPI method produces an infinite delay because zero EV halts the efficiency-based forecast entirely, signalling a complete work stoppage. The ES result, potentially resulting in over 15 days of accumulated delay, similarly confirms that the project is substantially behind schedule. Together, these indicators underscore the urgency of immediate recovery action for S2, while S1 remains

Table 16
ESM matrix for days 33 To 37.

Day	Project	EV	ES	I	SVc	SPIc
33	S1	35,530	31.977	−1.023	−1.023	0.969
	S2	0	0	−33	−33	0
34	S1	36,355	32.724	−1.276	−1.276	0.962
	S2	0	0	−34	−34	0
35	S1	37,184	33.456	−1.544	−1.544	0.956
	S2	0	0	−35	−35	0
36	S1	37,184	33.456	−2.544	−2.544	0.929
	S2	0	0	−36	−36	0
37	S1	37,184	33.456	−3.544	−3.544	0.904
	S2	0	0	−37	−37	0

Table 17
Time extension using ES metric (SPI).

On Day 33	Planned ES (X)	Actual ES (Y)	TS (Days) = X - Y	Additional Delay (Days)	Result
S1	33	31.977	33–31.977 = 1.023	1.023	1.023 days additional delay
S2	33	0.000	>15	>15	Over 15 days of additional delay

$$Time\ Extension\ (TE) = \frac{Planned\ Remaining\ Duration\ (Total\ planned\ days - Current\ day)}{SPI}$$

manageable with timely corrective attention.

However, as shown in Figure 16, the trend analysis of Time Extension (TE) from days 33 to 90 reveals fundamentally different delay patterns between Projects S1 (Blue line) and S2 (orange line). Project S1 exhibits generally low TE values in the early period (1.023–3.544 days, from days 33 to 41), interspersed with sharp but isolated spikes (most notably on days 39, 74, and 86), each triggered by temporary work stoppages where the EV drops to zero. These interruptions, although disruptive, are followed by recovery periods in which TE returns to zero (days 42–73), indicating that S1 intermittently regains schedule advantage when progress resumes. In contrast, Project S2 demonstrates a far more persistent and severe delay pattern: TE begins over 15 days on day 33 and remains elevated throughout, fluctuating between approximately 5.6 and 30 days, reflecting chronic work stoppages and sustained zero-EV periods. The consistently high TE trajectory for S2 signals entrenched schedule failure requiring immediate corrective actions such as resource reallocation or programme restructuring. By comparison, S1's intermittent spikes suggest isolated breakdowns that remain manageable if addressed promptly. Overall, the TE profiles indicate that while S1 experiences episodic disruptions with recovery potential, S2 suffers from continuous performance deterioration, which demands urgent intervention to prevent further schedule overrun.

III. Cost and Schedule Forecasting (EAC and TEAC)

As illustrated in Fig. 17, the absence of daily reports for project KIT between days 33 and 37 precluded the application of earned value forecasting techniques. Consequently, the quantification of potential cost and schedule outcomes was only performed for project GHS, utilising the RealCONs framework (Table 18).

EAC suggests costs will remain within budget, indicating minimal financial risk. However, on day 37 (Schedule Performance Index, SPI = 0.90), the Time Estimate at Completion (TEAC) forecasts a total duration of 100 days, a 10-day overrun. This implies:

Table 18
Cost and schedule forecasting for RealCONs and existing systems.

Projects	CPI: Day 37	EAC = $100,000/CPI	SPI: Day 37	TEAC = 90 Days/SPI
GHS (RealCONs)	1.03	$100,000/1.03 = $97,087	0.90	100 Days
KIT (Existing)	–	–	–	–

$$Estimate\ at\ Completion\ (EAC) = \frac{BAC}{CPI}\ \textbf{and}\ Time\ Estimate\ at\ Completion\ (TEAC) = \frac{Planned\ Duration}{SPI}$$

- **For GHS (S1):** Cost efficiency (CPI ≈ 1.03) masked schedule slippage (SPI fell from 0.97 to 0.90). The TEAC of 100 days implies 11.1% schedule overrun, necessitating proactive recovery measures.
- **For KIT (S2):** A complete work stoppage occurred between days 33 and 37 (EV = $0, AC = $1483/day). Consequently, EAC and TEAC are incalculable. It means the existing systems' EV fails, so it requires a root-cause analysis (e.g., resource levelling, scope change) and urgent corrective actions, such as re-baselining or resource levelling, to restore a positive SPI.

IV. CPI and SPI Behaviour: Days 33–37

The performance of the two projects diverged significantly. For GHS (S1), a resilient cost performance was maintained, with a stable CPI of 1.03 demonstrating that expenditure growth was contained relative to value earned (Fig. 18). However, this masked a steady erosion of the schedule, where a monotonic decline in the SPI from 0.97 to 0.90 indicated progressively worsening delays, likely stemming from unresolved operational bottlenecks. In stark contrast, KIT (S2) experienced a complete performance breakdown, characterised by a catastrophic halt in progress. This is evidenced by a sustained SPI of zero for five consecutive days and an accumulation of actual costs to $55,637.54 against zero earned value, confirming a total cessation of work that existing EVM frameworks failed to capture, necessitating immediate root-cause analysis and corrective intervention.

V. Managerial implications

The integration of regression analysis with Estimate at Completion (EAC) and Time Estimate at Completion (TEAC) allows project managers to anticipate risks more effectively (Table 19). By forecasting threshold breaches in the SPI, managers can quantify likely impacts such as a 10-day delay, prioritise corrective actions and communicate with stakeholders using statistically validated evidence. In this way, earned value metrics move beyond simple performance reporting and become predictive tools for decision-making.

A notable advantage of the S1 (RealCONs) system lies in its economic efficiency. Unlike the S2 framework, which incurs considerable administrative overhead in gathering daily reports, S1 reduces the number of missing reports from 35 to just 5. This improvement lessens the reporting burden, cuts indirect costs, and strengthens cost

Table 19
Managerial benefits of applying statistical techniques to S1 data.

Technique	Analytical Insight	Managerial Benefit
Regression (SPI trend)	SPI declines by 0.017/day; threshold breach forecast	Early warning of schedule risk enables preventive rescheduling
EAC (Cost forecast)	CPI = 1.03, so EAC ≈ BAC (cost stability)	Confirms budget safety; allows focus on schedule recovery
TEAC (Time forecast)	TEAC = 111% of baseline and 10-day delay projected	Provides a quantified estimate of delay; supports realistic replanning
	CPI stable, SPI variability = special cause	Confirms signals are reliable; justifies corrective intervention

Table 20
RealCONs framework Vs current digital tools.

Digital Tool/ Framework	Real-Time Monitoring	Early Delay Detection	Life-cycle Integration	Cost-Schedule Linkage	Adaptability	Limitations
RealCONs Framework [81]	✓ High	✓ Proactive	✓ Full life cycle	✓ Direct, automated	✓ High	Requires stable internet, authentication protocols
LPS [84]	× Manual	× Reactive	× Execution-focused	× Weak	✓ Moderate	Relies on facilitator discipline, no real-time data
4D/5D BIM [85]	✓ Static, Visual	× Limited	✓ Design–Execution	✓ Static, Visual	✓ Moderate	Labour-intensive updates, no dynamic cost tracking
Digital Twin [86]	✓ Dynamic	✓ Conditional	✓ Strong	✓ High fidelity	× Low	High setup cost, technical expertise required
CPM [87]	× Static	× post-facto	✓ Partial (Planning, Execution)	× Indirect	× Low	Poor adaptability to site changes, no real-time feedback

performance, as reflected in its superior CPI.

5. Discussion

This paper tackled the persistent challenge of monitoring Electrical and Instrumentation (E&I) works in complex, high-security construction projects, where traditional reporting systems often fail to provide the real-time insights required for effective decision-making [27]. While advanced technologies such as RFID [18], laser scanning [14], and digital twins [83] show considerable promise, their high costs and technical demands frequently preclude large-scale deployment. The RealCONs framework was developed to provide a practical and operationally viable alternative by synergistically combining ubiquitous QR code-based data capture, mobile devices, and live cloud-based analytics. In contrast to existing planning and control methods such as the Last Planner System (LPS) [84], 4D/5D (BIM) [85], Digital Twin [86] and Critical Path Method (CPM) [87], RealCONs facilitates continuous, live tracking through dynamic dashboards directly linked to earned value metrics (SPI and CPI). This capability provides an early warning mechanism for delays and cost discrepancies, enabling a more agile response in fast-paced project environments.

The case study findings robustly demonstrated that data collected through the RealCONs framework (S2) was markedly more consistent and reliable than that derived from existing tracking system reporting (S1). The S2 dashboards showed stable CPI and SPI trends, whereas S1 data exhibited irregular updates and frequent gaps, undermining its reliability for proactive control. Furthermore, the Power BI visualisations were found to significantly improve communication across stakeholder groups by providing intuitive, zone-based, and task-specific clarity, thus enhancing collective situational awareness. As summarised in Table 20, a comparative analysis positions RealCONs favourably against established digital tools, highlighting its strengths in real-time monitoring, full life-cycle integration, and proactive delay detection.

This paper presents a field-tested and scalable solution for the real-time monitoring of major construction projects, with a focus on E&I works. The RealCONs framework bridges a critical gap between field execution and project controls by linking on-site data capture (via QR codes) with earned value metrics and interactive Power BI dashboards. This integration significantly enhances project visibility, facilitates proactive delay management, and supports data-driven decision-making. The contributions of this work are threefold: (1) it delivers a live, QR-enabled tracking system, validated on real-world projects; (2) it provides a practical, deployable framework aligned with industry workflows; and (3) it demonstrates the successful integration of Earned Value Management (EVM) within a real-time digital monitoring system. Another feature of RealCONs is its capability for daily site situation tracking, a practice not yet commonplace in the construction industry. Importantly, the framework is cost-effective, leveraging existing technologies through an integrated data-driven approach. It requires minimal staff training (only for the Mobile App) and can be implemented rapidly, with only API connections and authentication permissions requiring initial configuration.

Although the present paper did not incorporate Building Information Modelling (BIM) or system dynamics-based simulation, the RealCONs architecture provides a strong foundation for such integration. The framework was intentionally designed as a workflow-compatible, real-time data-capture system, and its structured, time-stamped, component-level records, including activity metadata and earned-value indicators, can be directly aligned with BIM object identifiers (e.g., GUIDs) and Work Breakdown Structure codes. This interoperability offers a clear pathway towards predictive and simulation-enabled construction control systems. Recent literature highlights the growing convergence between BIM and artificial intelligence, where machine learning, deep learning, and predictive analytics are used to support automated decision-making, pattern recognition, and project outcome prediction [88]. These methods have been applied to schedule forecasting, dynamic resource allocation, and progress pattern detection, demonstrating how AI can extend BIM beyond static representation towards intelligent, data-driven optimisation [89]. Digital twin research further reinforces the value of linking real-time field data with BIM environments to support simulation, "what-if" analysis, and predictive evaluation of future project states [90]. Such models integrate time-series data into BIM to enable continuous monitoring, time-based simulation, and early detection of deviations. Advances in interoperability standards (e.g., IFC) and everyday data environments (CDEs) have been identified as prerequisites for effective BIM–AI ecosystems [91], aligning closely with the structured data governance approach embedded within RealCONs. In practical terms, the RealCONs dataset could underpin AI-based schedule prediction models, such as machine-learning regressors or classifiers trained on historical progress patterns, and system dynamics models [92, 93] capable of simulating delay propagation, productivity changes, and interactivity dependencies. These techniques would extend RealCONs from a real-time monitoring platform into a predictive and evaluative system, capable of identifying emerging risks, forecasting outcomes, and assessing alternative intervention strategies. When integrated with BIM environments, such predictive outputs could be embedded within 4D/5D model visualisations, enhancing spatial understanding and decision-making.

Collectively, these studies suggest that linking RealCONs with BIM and AI represents a credible and valuable direction for future research. By positioning RealCONs as a structured data backbone within broader digital ecosystems, future developments could leverage system dynamics and AI to deliver advanced simulation, prediction, and evaluation capabilities, progressing towards fully integrated digital-twin-enabled project control.

6. Research limitations and solutions

Although RealCONs delivered tangible operational benefits, its deployment surfaced some key constraints:

- Connectivity dependency: The system's reliance on stable internet connectivity meant that authentication failures or weak signal areas disrupted real-time data synchronisation, directly affecting reporting continuity. This underscores the need for offline or hybrid (online/offline) data capture capabilities, particularly in remote or infrastructure-constrained sites.
- QR code accessibility: Despite BIM-informed placement, physical damage, suboptimal positioning (e.g., on vertical surfaces), or environmental exposure, QR codes occasionally became unscannable, reducing field efficiency and highlighting the importance of durable, ergonomically placed markers.
- The value of AC may not be the real cost, so focusing on the schedule approach is suggested.
- API reliability: Analysts noted vulnerability to API disconnections, which risked data flow integrity. Integrating passive data sources (e.g., automated sensor logs) alongside active QR-triggered inputs could enhance overall system robustness and continuity.

7. Conclusion and future research

This paper presents and validates RealCONs, a real-time, QR-enabled monitoring framework applied to Electrical and Instrumentation (E&I) works. By combining mobile QR capture, a cloud-hosted SQL database, and dynamic Power BI dashboards, the framework addresses long-standing challenges of manual reporting, delayed feedback, and fragmented visibility. In practice, it delivered measurable improvements in project control, with more stable CPI and SPI trends and clearer insight into emerging delays, while maintaining continuity with established site workflows. The framework advances construction informatics by linking BIM-based asset localisation with real-time analytics and integrated visual dashboards, thereby bridging the persistent gap between on-site execution and digital oversight in coordination-intensive settings. To assess these improvements, the paper employed both earned value (EV) and earned schedule (ES) metrics.

EV was used because it remains the industry's standard reference for performance reporting and provides a familiar baseline for comparison [82]. However, its schedule indicators (CPI, SPI) are known to lose sensitivity as a project progresses and are influenced by the limitations of AC. ES was therefore adopted to overcome these issues. By expressing progress in time units, ES produces schedule-focused indicators (SVc, SPIc) that remain valid and responsive throughout the whole project duration [82,92]. ES also enables EV data to be translated into duration-based measures, improving the evaluation of schedule performance and supporting more accurate forecasts of completion time. Using EV alongside ES therefore demonstrates both continuity with conventional practice and the enhanced early delay detection achieved through RealCONs' real-time data capture. RealCONs proved effective in managing critical-path activities within high-security building projects, with electrical installations providing a suitable test case. The framework's principles are adaptable and can be transferred to civil, structural, or other coordination-heavy domains. Its design, based on commonly available technologies and streamlined workflows, supports broad deployment across diverse project environments. Early implementation also showed that intuitive interfaces and targeted user training promote rapid adoption, reflecting a positive cultural shift towards real-time, data-driven management. Looking ahead, further research should explore AI-driven analytics, enhanced offline capability for low-connectivity sites, and deeper integration with enterprise platforms. These developments could evolve RealCONs from a monitoring tool into a proactive decision-support system. Overall, RealCONs strengthens monitoring fidelity, supports timely managerial intervention, and provides a scalable foundation for future innovation in real-time project control.

Looking ahead, a logical avenue for development lies in linking RealCONs with Building Information Modelling (BIM), artificial intelligence (AI), and system dynamics-based simulation. Although not implemented in the present paper, the underlying data architecture of RealCONs provides a robust foundation for such integration. Structured progress records offer a suitable input for:

- AI-driven prediction, where historical time-series data from RealCONs could train machine-learning models to forecast slippage and productivity changes;
- system-dynamics simulation, using RealCONs data to calibrate causal-loop or stock-and-flow models to evaluate the propagation of delays and resource constraints;
- real-time BIM synchronisation, automatically updating 4D sequences with actual progress states to enhance visual cognition and communication; and
- scenario evaluation within BIM, enabling the simulation of alternative sequencing or resource-allocation strategies using live field data.

Exploring these linkages would extend RealCONs from a real-time monitoring tool to a predictive decision-support environment, combining digital capture, simulation, and advanced analytics within an end-to-end project control ecosystem. Fig. 19 conceptually illustrates how RealCONs provides the foundational data layer upon which BIM–AI analytics can be integrated to support simulation, prediction and enhanced visualisation.

CRediT authorship contribution statement

Kambiz Radman: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Mostafa Babaeian Jelodar:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Ruggiero Lovreglio:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.autcon.2026.106781.

Data availability

The data that has been used is confidential.

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